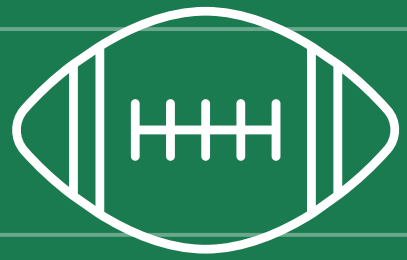
AI-NATIVE FOOTBALL INTELLIGENCE

Project Development & Pilot Partnership Proposal

An AI platform designed to analyze professional American football film, understand team systems and support tactical dialogue.



Technical Leadership:

CTO

Football Leadership:

Head of Football

02 PROJECT OBJECTIVE

Build a football intelligence platform that transforms game film into structured knowledge and actionable coaching support.

1. Automatically analyze individual plays and generate breakdown data.
2. Reconstruct plays through diagrams linked to the original footage.
3. Understand both the opponent and the team’s own football system.
4. Generate recommendations grounded in playbooks, personnel and game situations.
5. Support ongoing conversations with analysts, coaches and players.

> **The intended product is a persistent AI working environment, with documents and presentations generated from its underlying knowledge.**

Football preparation combines repetitive data work with demanding expert interpretation.



Analysts manually identify formations, concepts, fronts, coverages and outcomes.



Coaches review film to understand assignments, execution and tactical intent.



Staff reconcile different terminology and inconsistent labels.



Findings require further work before becoming reports, cut-ups or game plans.



New questions often require another search through previously reviewed footage.

This process consumes preparation time and makes analysis difficult to scale consistently across opponents and seasons.

04 THE PROPOSED WORKFLOW

The platform will connect film processing, football interpretation and coaching preparation within one system.

01	02	03	04	05	06
Import authorized game footage and available contextual information.	Segment plays and extract observable events.	Generate structured labels and reconstructed play diagrams.	Compare findings with opponent history and team knowledge.	Produce explanations, recommendations and preparation materials.	Incorporate coach corrections and finalize approved outputs.

Automation should progressively reduce manual intervention as individual capabilities meet validation standards.

The long-term objective is comprehensive football understanding with minimal routine human input.

Perception:	Identify visible players, alignments and movement.
Classification:	Recognize football structures and play characteristics.
Interpretation:	Evaluate relationships, likely assignments and execution.
Recommendation:	Suggest options relevant to the team’s system.
Communication:	Explain findings through dialogue, film and diagrams.

Each capability requires separate evaluation. Reliable formation recognition, for example, does not automatically establish reliable coverage interpretation or tactical advice.

06 VIDEO INGESTION & PLAY SEGMENTATION

The processing pipeline must convert raw footage into organized, searchable play sequences.

01

Ingest available sideline, end-zone or broadcast footage.

02

Detect play boundaries, pre-snap activity and post-play transitions.

03

Identify candidate snap and play-ending timestamps.

04

Synchronize multiple camera views when matching footage is available.

05

Attach game context from visible information or authorized supporting data.

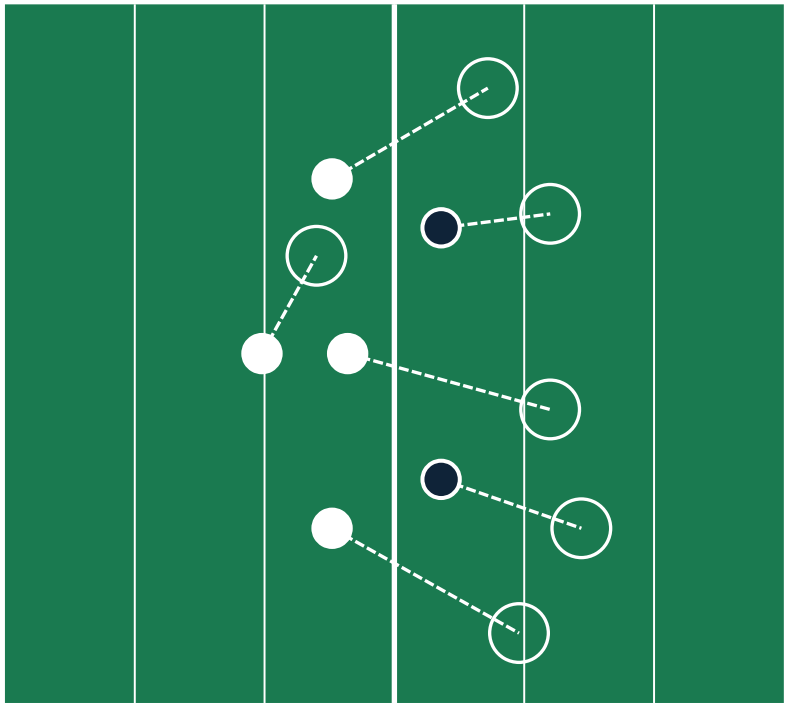
06

Flag missing frames, poor visibility and incomplete plays.

Film quality and camera coverage will determine which downstream outputs the system can support confidently.

The visual foundation requires spatial and temporal understanding of the field.

1. Use object detection and multi-object tracking to follow visible players.
2. Estimate field coordinates from yard lines, sidelines and other landmarks.
3. Compensate for camera movement, zoom and perspective changes.
4. Maintain player tracks through temporary occlusion where possible.
5. Associate tracks with team membership and available identity evidence.
6. Measure positional uncertainty and detect tracking failures.



Persistent tracking IDs and roster identities should remain separate until sufficient evidence supports a match.

A structured football ontology will define how the platform represents the game.

Game context:

Opponent, period, clock, score and possession.

Situation:

Down, distance, field position and personnel.

Pre-snap structure:

Formation, alignment, motion, front and shell.

Post-snap behavior:

Routes, blocking actions, rotation and pressure.

Outcome:

Gain or loss, completion, sack, turnover and other events.

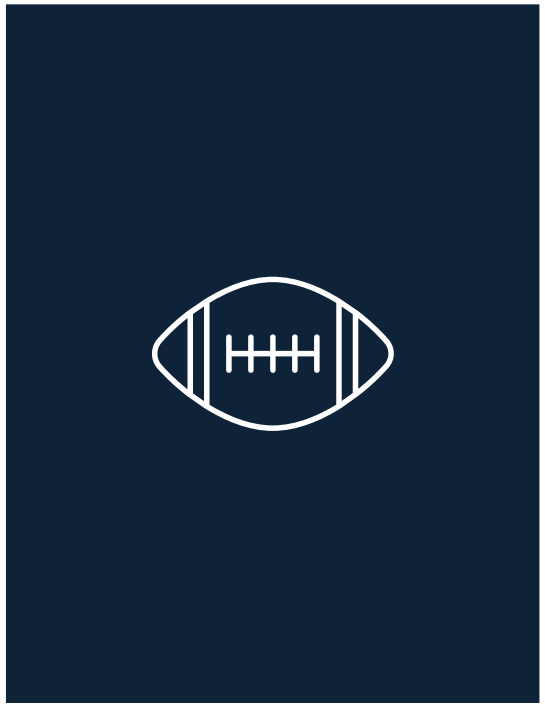
Evidence metadata:

Source clip, timestamps, confidence and review status.

A canonical terminology layer should support team-specific vocabulary without losing consistency across datasets.

09 OFFENSIVE ANALYSIS

The offensive model should connect pre-snap structure with post-snap execution.



Recognize formations, receiver distribution and backfield alignment.

Identify shifts, motions and changes in strength.

Classify run direction, blocking patterns and candidate run concepts.

Reconstruct routes and identify candidate passing combinations.

Evaluate protection behavior and responses to pressure.

Identify possible play-action, screen and RPO structures from available evidence.

Concept labels should express uncertainty when multiple calls could produce similar visible execution.

The defensive model should evaluate alignment, movement and responsibility relationships.



Identify fronts, defensive techniques, box counts and potential pressure threats.

Track secondary shells, rotations and changes in leverage.

Evaluate coverage hypotheses through defender movement and receiver relationships.

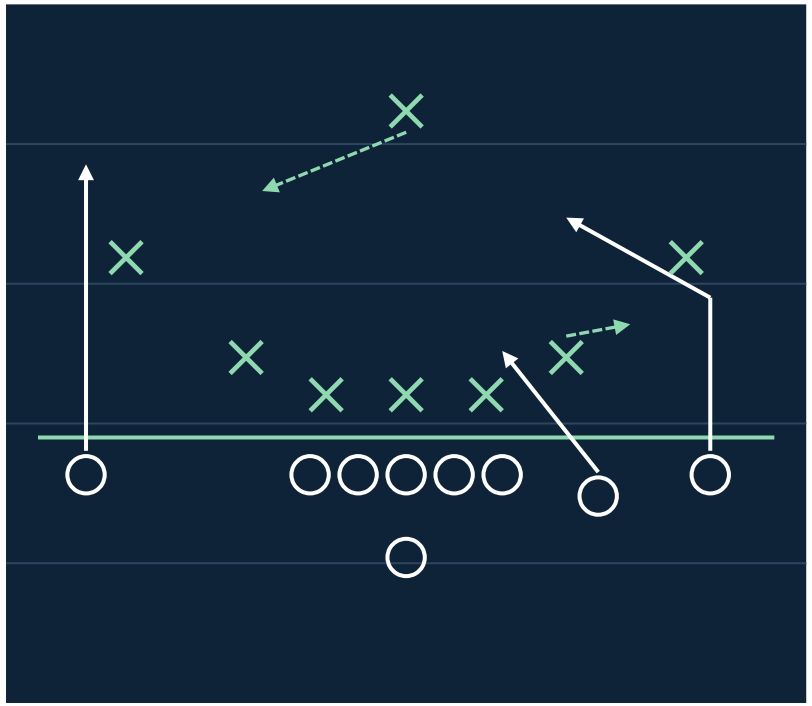
Distinguish rushers, simulated pressure, delayed pressure and dropping defenders.

Analyze visible run-fit behavior and pursuit.

Compare pre-snap presentation with post-snap structure.

Coverage identification must account for disguise, match rules and assignment ambiguity.

Play reconstruction should create an inspectable visual record of the action.



1. Map starting alignments onto a standardized field.
2. Draw observed movement paths and route trajectories.
3. Represent visible blocks, rush paths and defensive rotations.
4. Synchronize diagram playback with video timestamps.
5. Support alternate views and coach annotations.
6. Export selected diagrams into preparation materials.

Observed paths, inferred assignments and coach-confirmed responsibilities should use distinct visual conventions.

The reasoning layer should explain how football structures interact.

01 Evaluate route distribution against coverage rules and leverage.

02 Examine run-blocking structures against defensive fronts and box numbers.

03 Identify spacing conflicts and potential numerical advantages.

04 Analyze pressure paths against observed protection responses.

05 Compare successful and unsuccessful executions of similar concepts.

06 Generate alternative interpretations when the available evidence is incomplete.

> **Explanations should identify the supporting plays and distinguish tactical interpretation from directly observed facts.**

The platform needs a maintained representation of each team’s football system.

- Ingest authorized playbooks, installation materials and coaching notes.
- Map team terminology to the platform’s football ontology.
- Represent formations, calls, checks, adjustments and assignment rules.



- Track playbook versions and changes across preparation weeks.
- Retrieve relevant team concepts when answering tactical questions.
- Preserve coach-approved interpretations with their source and effective date.

Retrieval-augmented generation should ground responses in the applicable team material.

Tactical recommendations must consider who will execute them.

1. Maintain roster identities, positions and package roles.
2. Combine film observations with coach-validated player assessments.
3. Evaluate recurring technique patterns and execution tendencies.
4. Compare matchups within specific concepts and situations.
5. Adjust recommendations to the personnel available for preparation.
6. Separate measured performance from subjective evaluation.

Player assessments should remain contextual, with supporting clips and clear sample limitations.

The same analytical framework should support opponent preparation and internal evaluation.

Calculate tendencies by down and distance, field zone and formation.

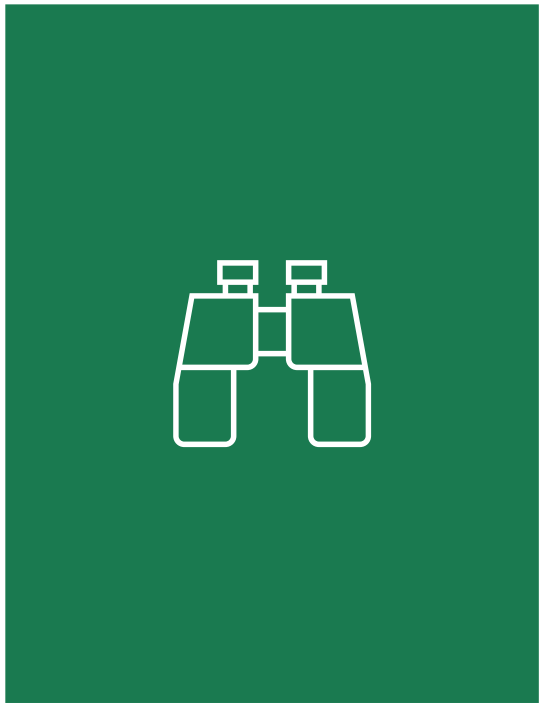
Examine personnel usage, motion patterns and situational calls.

Identify repeated sequences and potential offensive or defensive tells.

Compare behavior across games and against different opponents.

Highlight predictable patterns within the team’s own system.

Surface exceptions that challenge an apparent tendency.



Every statistical claim should expose its sample size, filters and underlying plays.

The conversational interface should allow users to investigate, challenge and refine analysis.

Example questions:

“Which pressures appear most often on third-and-medium?”

“Show the plays supporting this coverage interpretation.”

“Which concepts in our playbook fit this defensive look?”

“How does the recommendation change with a different personnel package?”

“What evidence contradicts this tendency?”



Responses should combine structured queries, retrieved team knowledge and relevant footage. Follow-up questions should retain the active football context.

Recommendations should translate scouting findings into team-specific options.

1. Match opponent tendencies with available calls and adjustments.
2. Explain the expected advantage and the conditions required.
3. Identify execution demands and relevant personnel constraints.
4. Present alternatives for likely opponent responses.
5. Support situational preparation and practice priorities.
6. Generate coaching notes, cut-ups and presentation drafts.

The system should explain why an option fits, what could invalidate it and which evidence supports it.

ANALYST, COACH & PLAYER EXPERIENCE



Analysts

Inspect automated labels, resolve uncertain plays and explore patterns through searchable data.



Coaches

Challenge interpretations, compare tactical options and approve preparation materials.



Players

Review approved clips, role-specific explanations and relevant opponent tendencies.

Shared Workflow

Corrections and annotations remain connected to their original plays. Access controls determine which users can view team knowledge, edit analysis and publish finalized material.

The platform should separate perception, structured analysis and language-based reasoning.

Application layer:	Conversation, visualization and output generation.
Reasoning layer:	Multimodal interpretation and grounded recommendations.
Knowledge layer:	Play records, playbooks and personnel information.
Football modeling layer:	Event recognition and structured classifications.
Perception layer:	Detection, tracking and field calibration.
Media layer:	Video storage, timestamps and clip retrieval.

Statistical calculations should query structured data directly, with language models explaining the results.

The development process must support reliable iteration.

01

Establish expert-reviewed reference datasets and annotation guidelines.

02

Separate training, validation and test data by game to reduce leakage.

03

Benchmark individual tasks before evaluating the complete workflow.

04

Compare performance across film conditions and football situations.

05

Use active learning to prioritize uncertain or informative examples.

06

Version models, datasets and evaluation results for reproducibility.

Production corrections should enter a reviewed improvement process before influencing future model releases.

Trust requires visible evidence and controlled use of sensitive football information.

- Calibrate confidence estimates and define abstention thresholds.
- Maintain revision history for labels and coaching corrections.
- Distinguish observed facts, model inferences and coach-confirmed knowledge.
- Isolate team data and enforce role-based permissions.
- Link generated claims to clips, records or playbook passages.
- Define authorized use, retention and model-training permissions.

A team’s proprietary materials should remain within its agreed access and usage boundaries.

EXECUTIVE LEADERSHIP

CTO

- Own technical architecture and engineering execution.
- Lead the AI and software development team.
- Define technical benchmarks and release processes.
- Translate product requirements into development priorities.

Head of Football

- Own football methodology and interpretation standards.
- Lead coordinators and position specialists.
- Define football evaluation criteria and resolve labeling disagreements.
- Coordinate coaching feedback and pilot adoption.

Joint responsibility: **Product scope, roadmap priorities and readiness for team use.**

CONTRIBUTING TEAMS & WORKING MODEL

AI & Software Team

Develop perception models, data infrastructure, reasoning services and user-facing workflows.

Football Expert Team

Provide scheme knowledge, review reference data and test positional and coordinator-level outputs.

Collaboration Process

Convert football questions into specifications, test implementations against film and review failures together.

Both teams will initially contribute without salary, with the intention of future shareholding. Ownership terms, responsibilities and contribution expectations require agreement among participants.

The pilot requires sustained participation from a team competing in league games.

Proposed commitments:

- Authorized access to relevant footage and agreed team materials.
- A designated staff contact and participating football specialists.
- Regular review sessions and documented feedback.
- Use of validated features during weekly preparation.
- Baseline measurements and agreed evaluation procedures.
- Permission-based reference materials for future commercial discussions.

The partnership should define data rights, responsibilities and success criteria before testing begins.

The pilot should assess both technical accuracy and practical coaching value.

1. Benchmark the existing manual preparation workflow.
2. Evaluate selected AI capabilities on expert-reviewed games.
3. Test additional games that the development process has not used.
4. Introduce validated outputs into weekly preparation.
5. Record errors, review time, adoption and recommendation usefulness.
6. Review results jointly through the CTO and Head of Football.

Early testing should run alongside the existing process. Match outcomes can provide context but cannot independently establish the product's impact.

KPI FRAMEWORK

Technical Quality	Precision, recall and F1 for applicable classifications, tracking continuity, identity-switch rates and reconstruction error.
Operational Efficiency	Processing latency, time to usable breakdown and total human review time per game.
Football Quality	Expert agreement, assignment interpretation quality and recommendation relevance.
User Adoption	Active users, repeat usage and integration into preparation workflows.
Reliability	Unsupported-claim rate, confidence calibration and performance on unseen games.

Numerical targets should follow baseline assessment and agreement on pilot scope.

DEVELOPMENT & COMMERCIAL ROADMAP

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
Specification & Reference Data	**Validated Core Prototype**	**Team Intelligence**	**League Pilot**	**U.S. Market Readiness**
Define supported film, football terminology, initial tasks and evaluation datasets.	Develop selected breakdown fields, play reconstruction and searchable evidence.	Integrate playbooks, personnel context and tactical dialogue.	Measure real preparation value and refine the product with participating staff.	Prepare demonstrations, a permission-based reference case and customer discovery.

Progression should depend on agreed evidence and readiness criteria.

STRATEGIC NETWORK & APPROVAL REQUEST

Commercial and Investment Pathway

Use validated pilot results to engage Europe-based coaches with NFL and NCAA coaching backgrounds, gather product feedback and pursue introductions to U.S. customers and investors.

Requested Approval

1. Establish the CTO and Head of Football leadership structure.
2. Confirm participation in product development and league testing.
3. Agree on data access, expert involvement and evaluation responsibilities.
4. Review measured results before expanding commercial activity.

The immediate goal is a validated product and a committed pilot partner, creating the evidence needed for U.S. market entry and future investment.